ERICK GATICA CANDIA

Ph.D. Student in Mechanical Engineering | M.Sc. & B.S. in Chemical Engineering

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Ph.D. researcher at the University of Colorado Boulder specializing in optical diagnostics, laser-based absorption spectroscopy sensors for hydrogen and natural gas combustion, thermal-fluid modeling, and Dual Comb spectroscopy. Experienced in university teaching, process simulation, solar desalination systems, and engineering consulting.

EXPERIENCE

Graduate Research Assistant

University of Colorado Boulder

2023 – Present | Boulder, CO

- Develop software and hardware for laser-based sensors applied to hydrogen engines and combustion systems in the Precision Laser Diagnostics for Energy and the Environment Lab under Dr. Greg Rieker
- Implement real-time data acquisition, optical diagnostics, and embedded system workflows

Research Assistant

University of Santiago of Chile

July 2022 – Present | Chile

- Research assistant in desalination processes using phase change materials and solar energy, focusing on building and modeling processes

Adjunct Lecturer / Instructor

University Autonomous of Chile

July 2022 – January 2023 | Chile

- Taught undergraduate courses in Process Control

University Instructor & Teaching Assistant

University of Santiago of Chile

Jan 2020 – Present | Chile

- Instructed laboratory and recitation classes in Transport Phenomena, Mass Transfer, Chemical Process Calculations, and Process Engineering Thermodynamics for Chemical Civil Engineering.
- Delivered mathematics, applied thermodynamics, and algebra recitations across the Faculty of Engineering and the Faculty of Administration and Economics.

Technical Editor

University of Santiago of Chile

May 2022 – July 2022 | Chile

- Reviewed and edited theoretical foundations and applied exercises for the *Handbook of Chemical Reactor Design* by Dr. Aldo Saavedra.

EDUCATION

Ph.D. in Mechanical Engineering

University of Colorado Boulder

2023 – 2028 | Boulder, CO

- Fourth-year Ph.D. student and Fulbright Scholar.
- M.Sc. in Mechanical Engineering (Conferred May 2025).

Fulbright Pre-Academic Program

Drexel University

2023 | Philadelphia, PA

- 6-week intensive multicultural academic leadership and research training program.

M.Sc. in Engineering Sciences, Chemical Engineering

University of Santiago of Chile

2019 – 2020 | Santiago, Chile

- GPA: 6.8 / 7.0 (Graduated with Maximum Distinction).

Chemical Civil Engineer (B.S. + M.Eng. equivalent)

University of Santiago of Chile

2015 – 2020 | Santiago, Chile

- Class Rank: 1st (Final Mark: 6.5 / 7.0, Graduated with Distinction).

SKILLS & LANGUAGES

Programming & Technical Computing

- Python, C++, MATLAB, LaTeX
- Data acquisition, signal analysis, and embedded system software.

Engineering Simulation & Modeling

- ANSYS Fluent, OpenFOAM, Aspen HYSYS, MS Excel
- CFD, finite volume modeling, and chemical process simulation.

Languages

- Spanish (Native)
- English (Fluent – TOEFL iBT: 94)

EXPERIENCE (CONTINUED)

Independent Consulting Engineer

Social Renewable

October 2021 – February 2022 Chile

- Provided consulting engineering services for client projects focused on environmental sustainability, energy efficiency, and renewable energy.

FONDECYT Support Staff

University of Santiago of Chile

October 2021 – April 2022 Chile

- Managed experiments and supervised thesis work in a FONDECYT project related to solar energy accumulation, spouted beds, and water desalination.

Chemical Civil Engineering Intern

Maltexco-Patagonia Malt

February 2020 – March 2020 Chile

- Integrated with the plant and laboratory teams to execute experimental designs for process optimization.
- Performed analysis of reducing sugars and correlated findings with the degree of caramelization in special caramel-type malts produced in toasters.

Fluid Mechanics Laboratory Assistant

University of Santiago of Chile

June 2018 – November 2020 Chile

- Conducted fluid mechanics experiments using prototype-scale equipment.

Differential Equations Teaching Assistant

University of Santiago of Chile

January 2017 – December 2018 Chile

- Instructed face-to-face recitation classes and supported engineering coursework in differential equations.

Chemical Laboratory Technician Intern

Eurofins GCL

November 2014 – Feb 2015 Chile

- Performed fertilizer analysis (determination of blue and yellow phosphorus, sulfur, and biuret) and nutritional analysis of food samples
- Prepared validated technical reports for national and international clients

VOLUNTEERING & LEADERSHIP

Member of SOAR Board, ISSS

CU Boulder

April 2026 – Present US

- Member of the Students Organizing for Arrival and Resources (SOAR) Board.
- Organized orientation activities for international students and produced marketing content.

PUBLICATIONS

High-speed thermal imaging of disk-shaped firebrands

Fire Safety Journal, Vol. 162 (2026)

Real-time EGR rate sensor for natural gas engine control

Proc. Combust. Inst., 105692 (2024)

Desalination using solar energy and phase change materials

J. Environ. Chem. Eng., Vol. 11 (2023)

Sawdust drying modeling with solar energy and PCMs

J. Energy Storage, Vol. 51 (2022)

Sawdust drying in spouted beds using solar energy

J. Energy Storage, Vol. 48 (2022)

COURSES & PROJECTS

Green Hydrogen: Opportunities and Applications

CORFO

Jan 2022 – Mar 2022

- Production, storage, industrial applications, and regulations.

CAE Fundamentals (ANSYS)

iESSS

Jun 2021

- Finite element and CFD analysis fundamentals.

Advanced Graduate Coursework

USACH

2019 – 2020

- Transport Phenomena, Thermodynamics, Control, Finite Volumes.

FONDECYT Research Project Thesis

USACH

2019 – 2021

- Modeling and experimental solar drying in spouted beds.

Process Control Loops Workshop

Binational Congress

Oct 2019

Chem-E-Car Project & Mining Course

ChemEng Congress

Oct 2018